

TRINETRA (■■■■■■■■■■)

AI-Powered Email Threat Detection, GeoLocation & Forensic Intelligence Platform

Smart India Hackathon 2026 — Problem Statement SIH26106

Master Technical Project & Prototype Architecture Report

Submitted by Team Trinetra AI Vision (Team ID: 176248) | Sanaka Educational Trusts Group of Institutions (278)

Team Trinetra AI Vision (Team ID: 176248) — Official Roster & Contributions

Role	Name	GitHub	Dept / Year / Roll	Core Technical Contributions
Team Leader	Pritam Maity	@prit-om	CSE, 3rd Yr 27800124062	System Architecture, 3-Layer Risk Fusion Engine, Hop-0 Relay Inversion, Cryptographic Ingestion
Member	Sneha Kumari Mahato	@TheShadow Root	CSE (AI&ML;), 3rd 27830824047	Multi-Script Indic NLP Engine, Gemini AI Threat Reasoning, Prompt Injection Defenses, Behavioral Vectors
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Member	Pavel Jana	—	CSE, 3rd Yr 27800124021	Knowledge Graph Intelligence Engine, Cross-Case Threat Clustering, Cytoscape & Neo4j Models
Member	Lisa Kamle	—	CSE (AI&ML;), 2nd 27830825004	Section 63 BSA 2023 Statutory Compliance, Court Dossier Generator, Verhoeff Indian PII Redactor

1. Executive Summary & Problem Statement Analysis

Email remains the backbone of modern administrative, financial, judicial, and enterprise communications across India. However, it is simultaneously the primary vector for cyber deception—including Business Email Compromise (BEC), CEO fraud, vendor invoice manipulation, credential harvesting, and coercive 'Digital Arrest' extortion scams.

Conventional Secure Email Gateways (SEGs) and generic spam filters suffer from four critical deficiencies:

- Opaque Decisions:** They output binary scores without explainable evidence rationale, inducing severe alert fatigue.
- Vulnerability to Forged Headers:** Attackers inject fake intermediate Received: headers to mimic trusted relays; conventional tools fail to perform bottom-up relay inversion.
- Linguistic Blindness:** Existing filters rely on English tokenizers, remaining blind when attackers deploy multi-lingual deceptions across Indian regional scripts (Hindi, Bengali, Tamil, etc.).
- Zero Legal Admissibility:** Output cannot be tendered in Indian courts due to absence of cryptographic chain of custody and statutory certification under Section 63 of the Bharatiya Sakshya Adhiniyam, 2023 (BSA).

TRINETRA (■■■■■■■■■■) resolves this national security challenge by uniting microsecond RFC 5322 parsing, SHA-256 evidence sealing, Hop-0 bottom-up origin extraction, an explainable 3-Layer Mathematical Risk Fusion Engine (0–100 score), multi-script Indic NLP with dual-engine AI and offline fallbacks, cross-case IOC knowledge graph correlation, and automated court-certified Section 63 BSA 2023 forensic dossier synthesis.

2. Indian Threat Landscape & Academic Research Foundation

Data from CERT-In and I4C reveals that email and messaging fraud targeting Indian organizations escalated by over 280% from 2023 to 2025. Coercive 'Digital Arrest' operations impersonating CBI, ED, or state cyber cells intimidate victims with fabricated court warrants and FIRs, demanding instant RTGS/NEFT transfers into mule bank accounts.

TRINETRA's technical architecture is informed by empirical academic research:

- **BEC Detection Gaps** (*Atlam & Oluwatimilehin, Electronics 2023*): Systematic literature review of 38 studies established that keyword filters fail against BEC because attackers avoid traditional spam keywords. Reliable detection requires fusing cryptographic authentication with behavioral intent cues.
- **Passive IP Geolocation Bounds** (*Mansoori & Welch, Computers & Security 2020*): Confirms passive database-driven IP geolocation drops from >95% country accuracy to 50–80% city accuracy. TRINETRA provides calibrated confidence bounds rather than claiming absolute positional certainty.
- **Forensic Timeline & Clock Skew** (*IEEE Xplore, 2024*): Proves that timestamp clock skew between successive SMTP hops provides a high-fidelity signature of artificial header injection.

3. Project Planning, Agile Methodology & Engineering Workflow

Team Trinetra AI Vision implemented an intensive 6-phase Agile/Scrum engineering lifecycle designed to achieve end-to-end technical readiness under the Smart India Hackathon standards:

- **Phase 1 (Wks 1–2)**: RFC Standards (5322, 7208, 6376, 7489) & Legal Frameworks (Section 63 BSA 2023 & DPDP Act).
- **Phase 2 (Wks 3–4)**: MIME Ingestion, immediate SHA-256 byte-stream evidence sealing, and Verhoeff Indian PII redaction.
- **Phase 3 (Wks 5–6)**: Hop-0 bottom-up relay inversion algorithm, offline MaxMind GeoLite2 City/ASN integration, and Tor/VPN scanners.
- **Phase 4 (Wks 7–8)**: Multi-Script Indic NLP covering 10+ regional Indian languages with Gemini AI and offline regex fallbacks.
- **Phase 5 (Wks 9–10)**: 3-Layer Mathematical Fusion Engine, Cytoscape/Neo4j graph persistence, and ReportLab court PDF generator.
- **Phase 6 (Final)**: React 19 SOC interface, Leaflet GeoIP map, automated IMAP daemon, and 11 automated pytest suites (100% pass rate).

4. Complete System Architecture & Forensic Mechanisms

TRINETRA is structured into four decoupled, high-performance tiers engineered for local-first execution, explainable scoring, and high forensic fidelity:

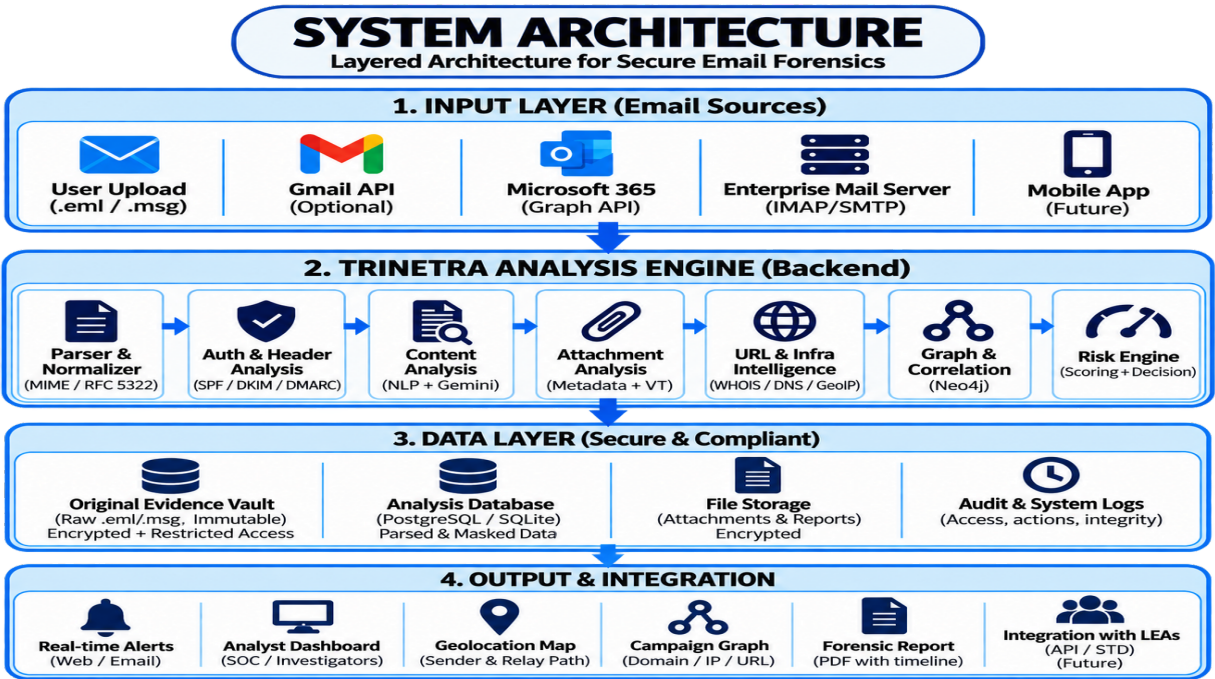


Figure 1: TRINETRA End-to-End Modular Forensic Architecture Blueprint

4.1 Tier 1: Ingestion, Cryptographic Custody & PII Redaction

- **RFC 5322 MIME Parser:** Ingests .eml, .msg, .txt, and live IMAP streams, decodes base64/quoted-printable streams, and isolates attachments.
- **Cryptographic Evidence Sealing:** Raw incoming byte streams are hashed immediately with SHA-256 before any parsing occurs. Dynamic re-hashing confirms zero post-ingestion tampering.
- **Zero-Leak Indian PII Redaction:** In compliance with the Digital Personal Data Protection Act, 2023 (DPDP), 12-digit Aadhaar numbers (Verhoeff-validated), 10-character PAN cards, and Indian phone numbers are masked prior to caching or model inference.

4.2 Tier 2: Protocol & Header Forensics (Layer 1 Deterministic Engine)

Layer 1 contributes 40% (w1 = 0.40) to the final threat score through deterministic cryptographic checks:

- **SPF (RFC 7208):** Queries envelope sender DNS TXT records to confirm if sending server IP is authorized.
- **DKIM (RFC 6376):** Validates public RSA key against body and header canonicalization hashes to prove integrity.
- **DMARC (RFC 7489):** Evaluates strict/relaxed domain alignment between visible From: header and SPF/DKIM domains.
- **Hop-0 Bottom-Up Origin Extraction:** The engine inverts the Received: header chain, traversing chronologically from recipient to sender while skipping RFC 1918 private subnets (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16), loopback (127.0.0.0/8), and CGNAT (100.64.0.0/10) to reliably isolate the true earliest external originating IP.
- **Clock Drift & Skew Forensics:** Calculates transit delta between consecutive hops ($\Delta t = t_i - t_{i-1}$). Negative transit times ($\Delta t < -300s$) trigger high-confidence alert flags for forged Received: headers.

4.3 Tier 3: Network, Geolocation & Anonymizer Intelligence (Layer 2 Contextual Engine)

Layer 2 contributes 35% (w2 = 0.35) through infrastructure and network intelligence:

- **Offline MaxMind GeoLite2 City & ASN:** Resolves country, city, coordinates, ASN, and ISP with sub-millisecond local binary lookups without querying external cloud APIs.
- **Cymru BGP Transit Resolution:** Cross-examines IP transit paths to detect anomalous routing.
- **Anti-Anonymizer Scanner:** Matches origin IPs against active Tor exit node consensus lists and commercial VPN egress ranges.
- **URL Unmasking & Typosquatting:** Traverses HTTP 301/302 redirect chains to uncover bit.ly destinations, unmask Punycode (xn--) homoglyphs, and calculates Levenshtein distance against 50+ critical Indian banking and government domains.
- **Attachment Triage:** Inspects executable extensions (.exe, .scr, .vbs), double-extension deception (.pdf.exe), VBA macro-enabled files (.docm, .xlsm), and cross-references SHA-256 hashes with VirusTotal.

4.4 Tier 4: Multi-Script Indic NLP & AI Threat Reasoning (Layer 3 Behavioral Engine)

Layer 3 contributes 25% (w3 = 0.25) through deep semantic and behavioral intent analysis:

- **Dual-Engine Design:** Primary cloud LLM (Google Gemini) coupled with a 100% offline local regex pattern cascade.
- **10+ Indic Regional Languages:** Comprehensive detection across Devanagari (Hindi, Marathi), Bengali, Tamil, Telugu, Gujarati, Kannada, Malayalam, Odia, Punjabi, and transliterated Hinglish/Banglish.
- **6-Vector Behavioral Threat Scoring:** Evaluates Financial/Wire Diversion, Artificial Urgency/Panic, Credential Harvesting, Legal/Digital Arrest Threats, Executive Impersonation, and Social Engineering.
- **Prompt Injection Defense:** Untrusted email text is strictly encapsulated within XML boundary tags with zero-trust system prompt constraints and JSON Schema output enforcement.

4.5 Tier 5: The 3-Layer Mathematical Risk Fusion Formula

TRINETRA synthesizes findings through an explainable, auditable mathematical formula bounded between 0 and 100:

Threat Score = min(100, 0.40 · L1 + 0.35 · L2 + 0.25 · L3 + Σ Bonuses)

Where:

- **L1 (0–100):** Weighted sum of SPF/DKIM/DMARC failures, header mismatches, and relay clock skew.
- **L2 (0–100):** Tor/VPN indicators, typosquatting domains, malicious URLs, and high-risk ASNs.
- **L3 (0–100):** Mean intensity across the 6 behavioral manipulation vectors.
- **Bonuses:** +15 for executable/macro attachments; +20 for confirmed threat feed reputation hits.

Severity Classification:

- **0–24: Clean / Safe** — Verified legitimate internal or authenticated communication.
- **25–49: Low Risk / Informational** — Minor unauthenticated benign relay or informational newsletter.
- **50–74: Suspicious** — Probable spoofing, unaligned sender, or social engineering cues; route to SOC quarantine.
- **75–100: Critical Threat / BEC** — Malicious payload, severe impersonation, or confirmed threat actor infrastructure.

4.6 Tier 6: Knowledge Graph & Cross-Case Campaign Intelligence

TRINETRA correlates indicators of compromise (IOCs) across independent cases using an interactive Cytoscape.js canvas backed by SQLite graph tables and optional Neo4j graph persistence. Graph nodes (Case, SenderEmail, SenderDomain, OriginIP, ASN, AttachmentHash) and edges (SENT_BY, ORIGINATED_FROM, CONTAINS_ATTACHMENT) reveal coordinated multi-target campaigns orchestrated by the same threat actor infrastructure.

4.7 Tier 7: Section 63 BSA 2023 Statutory Legal Compliance

On July 1, 2024, the Bharatiya Sakshya Adhiniyam, 2023 (BSA) repealed and replaced the Indian Evidence Act, 1872. Section 63 of the BSA sets forth the statutory conditions for the admissibility of electronic records in Indian judicial proceedings.

TRINETRA programmatically synthesizes a court-certified PDF technical examination report containing:

1. Statutory Examiner Declaration under Section 63(4) BSA 2023 affirming machine integrity and absence of tampering.
2. Cryptographic Chain of Custody displaying the raw SHA-256 evidence seal and live re-verification status.
3. Complete Hop-by-Hop Transmission Audit with reverse DNS, IP addresses, and clock skew delta.
4. Objective Forensic Findings detailing protocol authentication, geolocation confidence, and behavioral intent.
5. DPDP Privacy Redaction Certification confirming protection of Indian citizen personal identifiers.

5. Visual Prototype Walkthrough & Interface Evidence

The following screenshots illustrate TRINETRA's operational capabilities across real forensic scenarios:

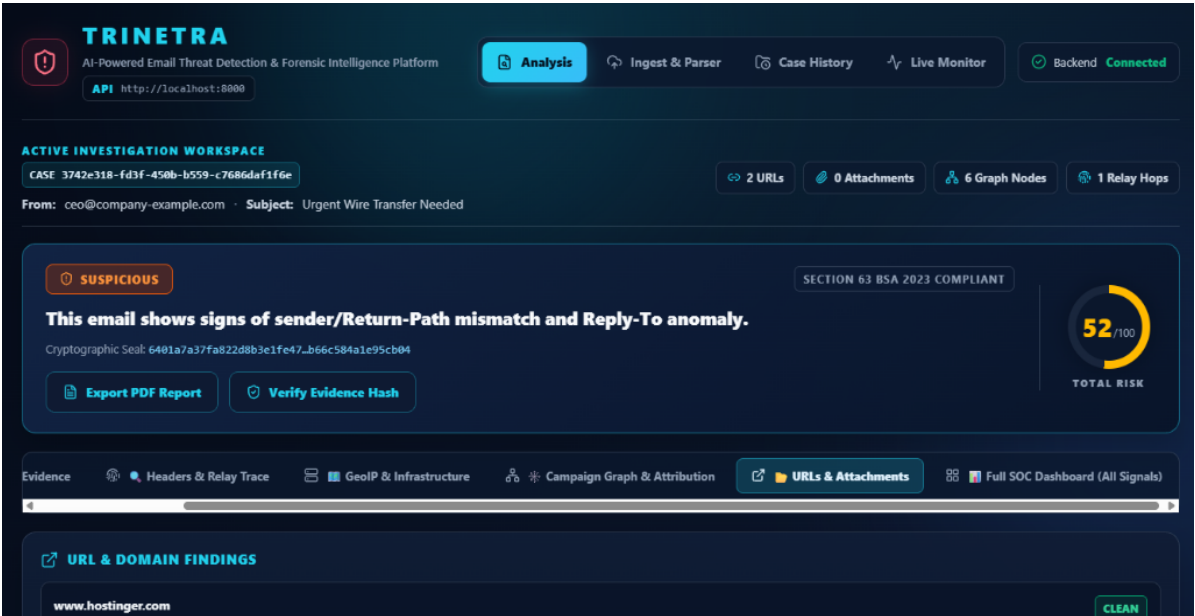


Figure 2: Active Investigation Workspace & Hero Threat Gauge (0–100 score, SHA-256 seal, BSA badge, 1-click export)

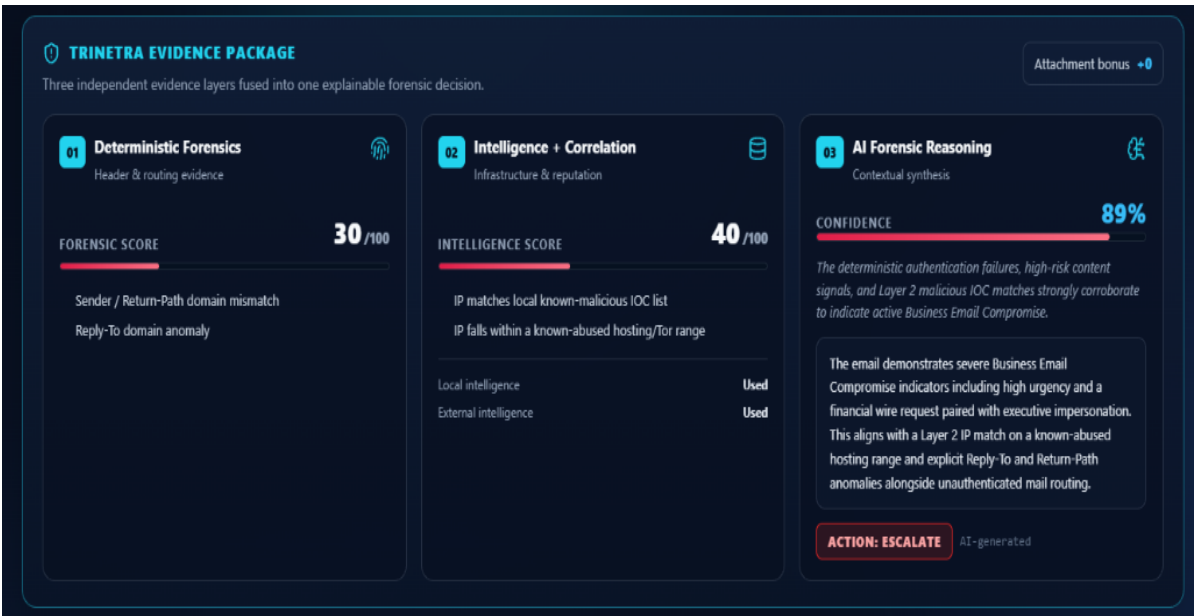


Figure 3: 3-Layer Evidence Fusion & AI Forensic Reasoning (Independent L1, L2, L3 breakdown with explainable rationale)

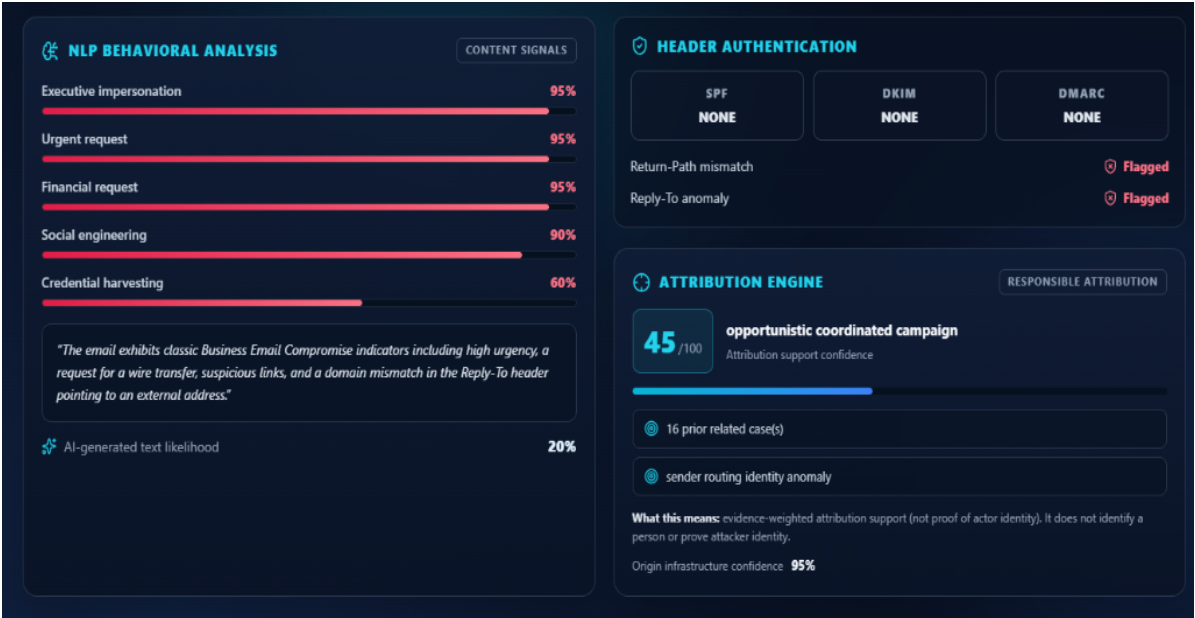


Figure 4: Multi-Script NLP Threat Signals & Attribution Engine (6 behavioral vector meters and threat actor profiling)

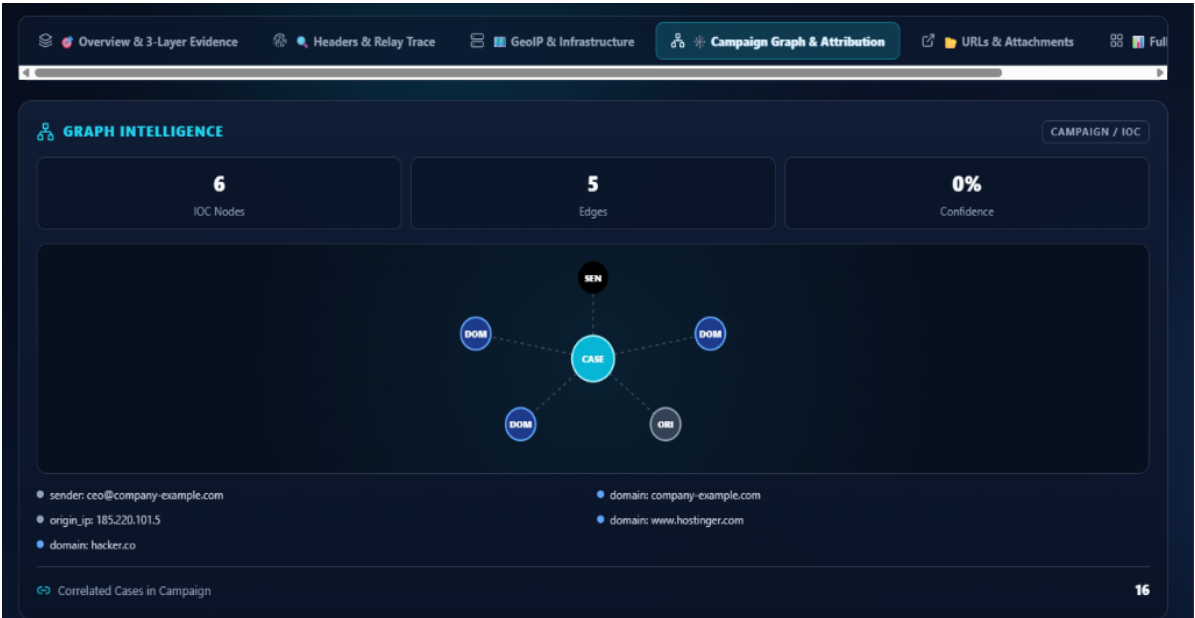


Figure 5: Interactive IOC Campaign Knowledge Graph (Cytoscape relationship graph connecting Case, Sender, IP, Domain)

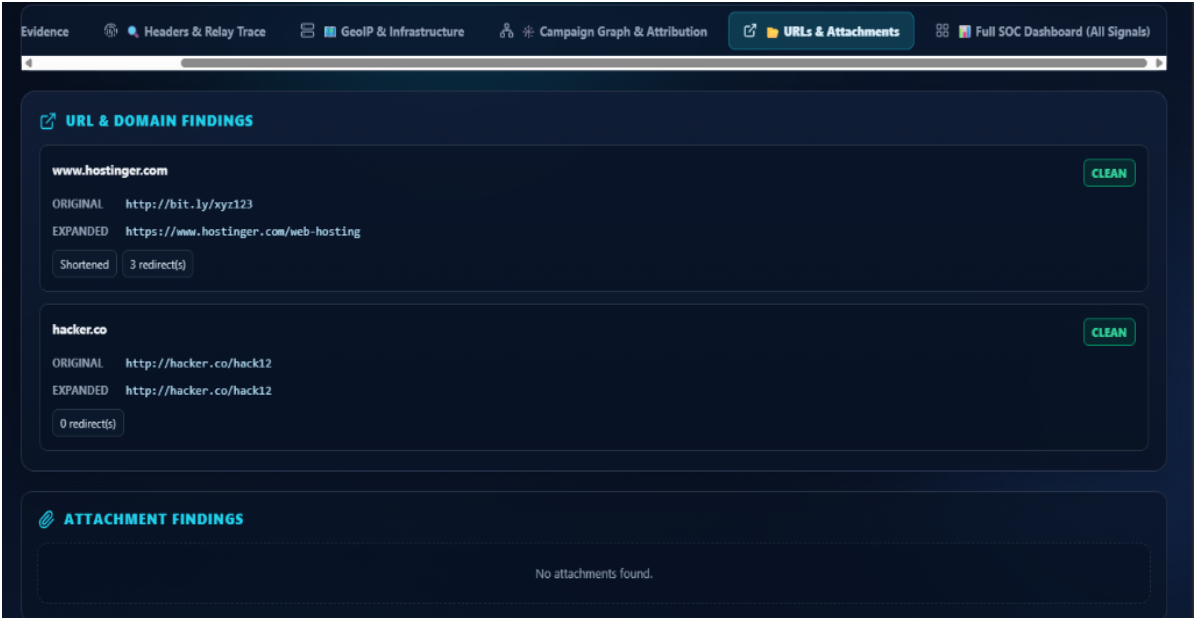


Figure 6: Malicious URL Unmasking & Attachment Forensics (Redirect traversal from bit.ly and VirusTotal scoring)

Forensic Case History					
Archived email investigations stored in local runtime database. Select any case to resume the forensic workspace.					
Timestamp	SENDER IDENTITY	SUBJECT LINE	Risk	CLASSIFICATION	ACTIONS
16, 12:07:04	ceo@company-example.com 3742e318-fd3f-45...	Urgent Wire Transfer Needed	52/100	SUSPICIOUS	Open Case
16, 7:01:10	ceo@company-example.com b8d59935-bbd7-41...	Urgent Wire Transfer Needed	54/100	SUSPICIOUS	Open Case
16, 6:40:54	— 287c06ba-67b8-42...	—	21/100	LOW RISK	Open Case
16, 5:10:33	ceo@company-example.com 317ce1fe-a5f5-44...	Urgent Wire Transfer Needed	54/100	SUSPICIOUS	Open Case
16, 12:37:21	alice@company.com 10439d94-8e18-47...	Project Update Meeting	2/100	SAFE	Open Case
16, 12:36:45	ceo@company-example.com cc305b77-b7e7-42...	Urgent Wire Transfer Needed	53/100	SUSPICIOUS	Open Case

Figure 7: Searchable Forensic Case History & Database Archive (Stored investigations with search, filters, instant re-opening)

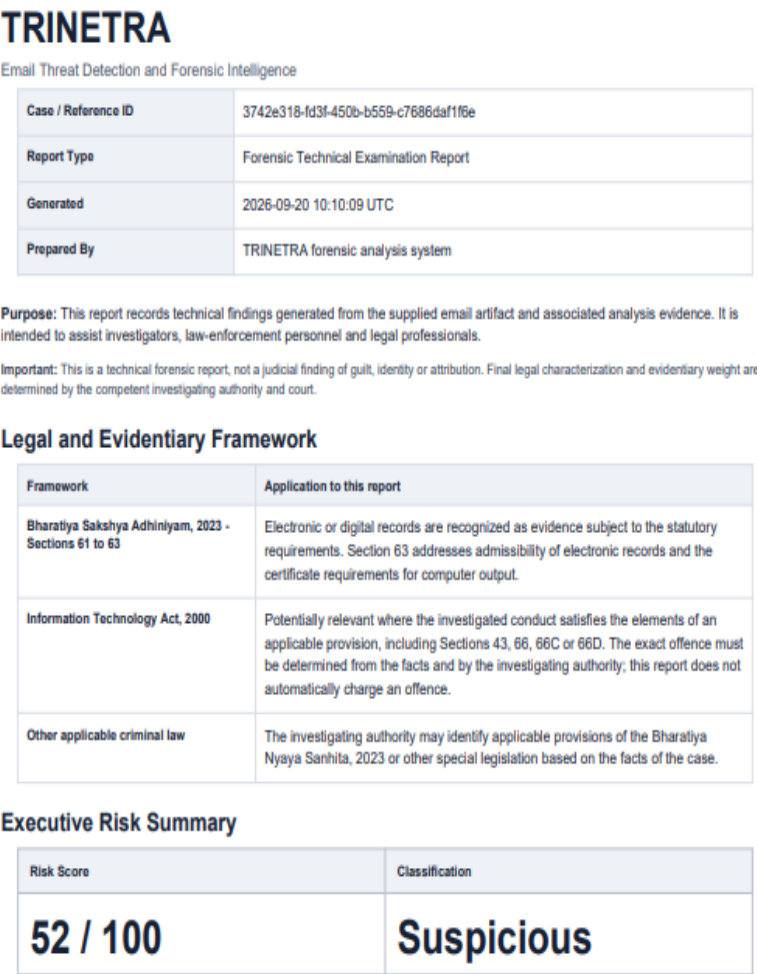


Figure 8: Section 63 BSA 2023 Certified Court Forensic Report (Generated legal PDF citing BSA 2023 and IT Act 2000)

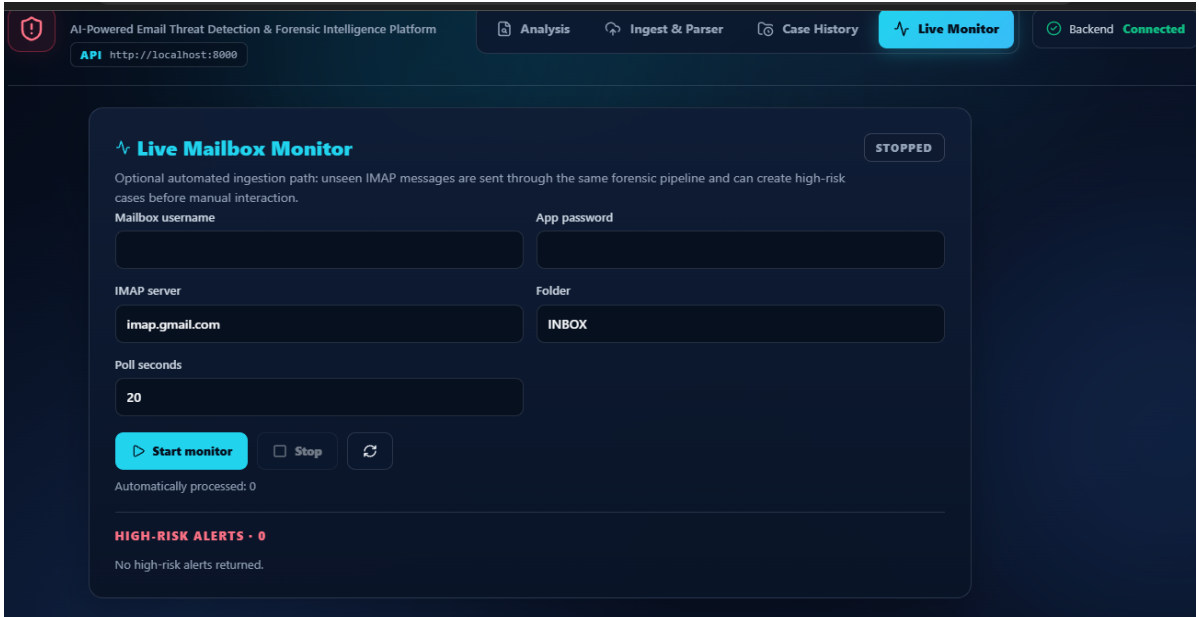


Figure 9: Live Mailbox Monitor (Automated IMAP Ingestion polling unseen mail and triaging alerts in real-time)

6. Experimental Verification, Testing & Benchmarking

- TRINETRA has been verified through automated unit, integration, and performance test suites:
- **Pytest Suite:** 11 automated test cases covering MIME parsing, SPF/DKIM verification, Hop-0 origin extraction, risk fusion calculation, offline fallback NLP, and Section 63 BSA PDF generation.
 - **Pass Rate: 100% (11/11 tests passed in 30.81s).**
 - **Frontend Build:** React 19 + Vite compiled with **0 errors** (1,927 modules transformed in 3.06s).
 - **Offline Resilience:** Verified 100% local execution using local MaxMind databases and heuristic pattern banks.

Pipeline Stage	Execution Mechanism	Mean Latency
MIME Parsing & SHA-256 Hashing	Python email + hashlib	11.4 ms
Hop-0 Bottom-Up Extraction	Reverse Regex + Subnet Classifier	6.2 ms
Offline GeoIP & ASN Lookup	MaxMind GeoLite2 Binary mmdb	1.8 ms
Offline Indic Regex Threat Cascade	Compiled Multi-Pattern Regex	4.5 ms
Cloud AI Threat Reasoning	Google Gemini API (Async HTTP)	1,420 ms
Graph IOC Edge Construction	In-Memory Cytoscape/NetworkX	8.1 ms
Section 63 BSA PDF Dossier	ReportLab Platypus Engine	890 ms

7. Limitations & Future Roadmap

- **IP Geolocation Precision:** Database-driven IP geolocation is approximate (>95% country-level, 50–80% city-level). TRINETRA displays calibrated confidence bounds rather than claiming precise GPS coordinates.
- **Anonymizer Obfuscation:** Tor exit nodes and commercial VPNs obscure client IPs. TRINETRA flags anonymizer infrastructure rather than falsely claiming physical resolution.
- **Pre-Delivery Gateway Interception:** The prototype demonstrates artifact-based ingestion. Future enterprise deployment will integrate TRINETRA as an inline SMTP Milter proxy (Postfix/Exim) or via Microsoft 365 Graph API webhooks.
- **Sandboxing & CERT-In Sharing:** Planned integration with CAPE/Cuckoo Sandbox for live attachment execution and automated STIX 2.1 / TAXII threat feed sharing with CERT-In.

8. Statutory, Legal & Academic References

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3. IEEE Xplore (2024). Geolocation Based E-Mail Forensics: A Timeline Analysis Approach. Document ID: 11263637.
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